

The Policy Gradient Theorem

Episodic Tasks: The Trajectory Route

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1 The Objective and Its Two Forms

This is a second derivation of the policy gradient theorem, by the *trajectory route*, complementing the recursive (Bellman) derivation of the companion report. Both reach the same theorem. This route is the one sampling algorithms (REINFORCE onward) are built on, and it is where the one genuinely subtle step, *causality*, appears and must be proved. The general intuition (what θ controls, why no transition derivative survives, how the visitation enters) was built in the companion report and is not repeated here; we keep only what is specific to this route.

1.1 Setup

The task is *episodic*: an absorbing terminal state, modeled as a zero-reward self-loop, ends every run. The terminal time T , the step of first entering that state, is a random hitting time, finite almost surely, so trajectories have different lengths. With $\theta \in \mathbb{R}^{d'}$ the interaction is

$$S_0 \sim \rho_0, \quad A_t \mid S_t = s \sim \pi_\theta(\cdot \mid s), \quad (S_{t+1}, R_{t+1}) \mid (S_t = s, A_t = a) \sim p(\cdot, \cdot \mid s, a),$$

and only π_θ carries θ . The return, value, and action-value are

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}, \quad V^{\pi_\theta}(s) = \mathbb{E}[G_t \mid S_t = s], \quad Q^{\pi_\theta}(s, a) = \mathbb{E}[G_t \mid S_t = s, A_t = a],$$

with $\gamma \leq 1$ ($\gamma = 1$ is fine, since T is finite the absorbing state's zero reward truncates the sum at T). $V^{\pi_\theta}, Q^{\pi_\theta}$ carry no time index: for a stationary policy the situation depends on the state, not the clock.

1.2 Two Forms of the Objective, and Why They Are the Same

We score a policy by its expected return from the start, and we will use both ways of writing it:

$$J(\theta) = \underbrace{\sum_{s_0} \rho_0(s_0) V^{\pi_\theta}(s_0)}_{\text{start-value form}} = \underbrace{\mathbb{E}_{\tau \sim p_\theta}[G_0]}_{\text{trajectory form}}.$$

They are the same number by one line of conditioning. Since $V^{\pi_\theta}(s_0) = \mathbb{E}[G_0 \mid S_0 = s_0]$, averaging over ρ_0 and collapsing the nested average (the tower property $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$) gives

$$\sum_{s_0} \rho_0(s_0) V^{\pi_\theta}(s_0) = \mathbb{E}_{S_0 \sim \rho_0}[\mathbb{E}[G_0 \mid S_0]] = \mathbb{E}_{\tau \sim p_\theta}[G_0].$$

The recursive route differentiates the left form (the value); the trajectory route differentiates the right form (the average return). Same objective, two handles.

2 From the Objective to a Score Times a Return

A trajectory is $\tau = (S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T, S_T)$, and its probability factorizes along the interaction:

$$p_\theta(\tau) = \rho_0(s_0) \prod_{t=0}^{T-1} \underbrace{\pi_\theta(a_t \mid s_t)}_{\text{depends on } \theta} \underbrace{p(s_{t+1}, r_{t+1} \mid s_t, a_t)}_{\text{no } \theta}.$$

We want $\nabla_\theta J = \nabla_\theta \mathbb{E}_{\tau \sim p_\theta}[G_0]$. The dependence on θ sits entirely in *how likely each trajectory is*, not in $G_0(\tau)$ (a fixed number once τ is fixed). The log-derivative identity $\nabla_\theta p_\theta = p_\theta \nabla_\theta \log p_\theta$ turns “differentiate the sampling probability” into “average against it” (interchange of ∇_θ and the trajectory sum is justified by the standing regularity: finite $\mathcal{S}, \mathcal{A}$, bounded reward, T finite a.s. with light tail, π_θ smooth):

$$\nabla_\theta J(\theta) = \sum_{\tau} \nabla_\theta p_\theta(\tau) G_0(\tau) = \mathbb{E}_{\tau \sim p_\theta} [G_0 \nabla_\theta \log p_\theta(\tau)].$$

Now log turns the product into a sum, and only the policy factors carry θ (the start and every environment factor differentiate to zero), so the trajectory score collapses to a sum of per-step policy scores:

$$\nabla_\theta \log p_\theta(\tau) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t | s_t), \quad \implies \quad \nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[G_0 \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t | S_t) \right].$$

This is exact but crude: every action’s score is multiplied by the *entire* episode return G_0 , including rewards collected *before* that action was taken. Removing those is the next step, and it is both the correct credit assignment and the first variance reduction.

3 Causality: Crediting Each Action With Its Future

Abbreviate the score at step t by $Z_t := \nabla_\theta \log \pi_\theta(A_t | S_t) \in \mathbb{R}^{d'}$, so $\nabla_\theta J = \sum_t \mathbb{E}_{\tau \sim p_\theta} [G_0 Z_t]$. Split the return at step t into past and future:

$$G_0 = \underbrace{\sum_{m=0}^{t-1} \gamma^m R_{m+1}}_{C_t \text{ (past)}} + \underbrace{\gamma^t \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}}_{G_t \text{ (future)}} = C_t + \gamma^t G_t.$$

C_t is the discounted reward already banked over steps $1, \dots, t$, before A_t is even chosen. The claim is that this past contributes nothing to the gradient:

$$\boxed{\mathbb{E}_{\tau \sim p_\theta} [C_t Z_t] = 0.}$$

3.1 The Proof, Conditioning on the History Before the Action

Let H_t be everything observed up to and including arrival at S_t , but before drawing A_t :

$$H_t := (S_0, A_0, R_1, S_1, \dots, R_t, S_t).$$

Average over everything given H_t , then over H_t (tower property): $\mathbb{E}[C_t Z_t] = \mathbb{E}_{H_t} [\mathbb{E}[C_t Z_t | H_t]]$. Once H_t is fixed:

- **Known:** $S_0, \dots, S_t, A_0, \dots, A_{t-1}, R_1, \dots, R_t$. In particular $C_t = \sum_{m=0}^{t-1} \gamma^m R_{m+1}$ is built only from $R_1, \dots, R_t \subset H_t$, so it is a fixed number.
- **Still random:** the action $A_t \sim \pi_\theta(\cdot | S_t)$, not yet drawn. The only randomness in $Z_t = \nabla_\theta \log \pi_\theta(A_t | S_t)$ is through A_t (S_t is known).

So C_t pulls out and the inner expectation is the average of the score over the one undrawn action:

$$\mathbb{E}[C_t Z_t \mid H_t] = C_t \mathbb{E}_{A_t \sim \pi_\theta(\cdot \mid S_t)} [\nabla_\theta \log \pi_\theta(A_t \mid S_t)] = C_t \cdot 0 = 0,$$

because that average is zero:

$$\mathbb{E}_{A_t \sim \pi_\theta(\cdot \mid S_t)} [\nabla_\theta \log \pi_\theta(A_t \mid S_t)] = \sum_a \pi_\theta(a \mid S_t) \nabla_\theta \log \pi_\theta(a \mid S_t) = \sum_a \nabla_\theta \pi_\theta(a \mid S_t) = \nabla_\theta 1 = 0.$$

Averaging over H_t gives $\mathbb{E}[C_t Z_t] = 0$. ■ Substituting $G_0 = C_t + \gamma^t G_t$ and dropping the vanished past at every t :

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[\sum_{t=0}^{T-1} \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t \mid S_t) \right].$$

The *reward-to-go* form: each action's score is weighted only by the return collected from that step onward.

Why the expected score is zero: the policy mass is conserved. At S_t the policy spreads a total probability of one across the actions, and it does so for *every* θ . Nudging θ only *redistributes* that mass, adding probability to some actions and taking the same amount from others, because the total is pinned at one. The score $\nabla_\theta \log \pi_\theta(a \mid S_t)$ is exactly the per-action rate at which mass flows as θ moves, and averaging it under the policy, $\sum_a \pi_\theta \nabla_\theta \log \pi_\theta = \sum_a \nabla_\theta \pi_\theta = \nabla_\theta \sum_a \pi_\theta$, adds up all those flows. They must net to zero, because the total they add to cannot change from one. That is all $\nabla_\theta 1 = 0$ is saying: mass moved in equals mass moved out.

The meaning, and a free variance reduction. The score asks “at S_t , make A_t more or less likely?” The rewards before t were collected before A_t was chosen, so changing the probability of A_t cannot reach back and alter them. The proof states this exactly: given H_t the past C_t is fixed, the only free quantity is A_t , and its score averages to zero, so past reward times present score averages to nothing. Beyond correctness, this is the first variance reduction and it is free: the removed piece $C_t Z_t$ has mean zero, so dropping it adds no bias, while C_t was real fluctuating reward uncorrelated with the decision, so carrying it only injected noise. The next variance reduction, subtracting a baseline $b(S_t)$ from G_t , uses this very same zero-mean identity and belongs to the REINFORCE report.

4 From the Sampled Return to the Action-Value

The reward-to-go G_t is one sampled number; replacing it by its average leaves the gradient unchanged, and that average is Q^{π_θ} . Condition now on the pair (S_t, A_t) . Given (S_t, A_t) the score Z_t and the discount γ^t are known, while the future rewards, hence G_t , are still random, so

$$\mathbb{E}[\gamma^t G_t Z_t \mid S_t, A_t] = \gamma^t Z_t \underbrace{\mathbb{E}[G_t \mid S_t, A_t]}_{= Q^{\pi_\theta}(S_t, A_t)} = \gamma^t Q^{\pi_\theta}(S_t, A_t) \nabla_\theta \log \pi_\theta(A_t \mid S_t).$$

Summing over t :

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim p_{\theta}} \left[\sum_{t=0}^{T-1} \gamma^t Q^{\pi_{\theta}}(S_t, A_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t) \right].$$

The two routes meet at $Q^{\pi_{\theta}}$, from opposite ends. The recursive route *started* from $Q^{\pi_{\theta}}$ (it sat inside $V^{\pi_{\theta}} = \sum_a \pi_{\theta} Q^{\pi_{\theta}}$) and unrolled the visitation around it. The trajectory route *arrives* at $Q^{\pi_{\theta}}$: it began from the raw sampled return G_0 , stripped the past by causality, and only now recovers $Q^{\pi_{\theta}}$ by averaging the sampled future. Same object, reached from opposite directions. This is the precise sense in which the two derivations are one theorem.

5 The Theorem

We turn the trajectory expectation into a sum over states in three steps. First, pull the time sum out by linearity:

$$\nabla_{\theta} J(\theta) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{\tau \sim p_{\theta}} [Q^{\pi_{\theta}}(S_t, A_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t)].$$

Second, reduce each term to its (S_t, A_t) marginal. The bracket depends on the whole trajectory only through the pair (S_t, A_t) , so only that pair's distribution matters, and that distribution factorizes as occupancy times policy, $\mathbb{P}(S_t=s, A_t=a) = \mathbb{P}(S_t=s) \pi_{\theta}(a | s)$:

$$\mathbb{E}_{\tau \sim p_{\theta}} [Q^{\pi_{\theta}}(S_t, A_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t)] = \sum_s \mathbb{P}(S_t=s) \sum_a \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s).$$

Third, swap the order of summation: the inner (s, a) sum is the same at every t apart from the weight $\mathbb{P}(S_t=s)$, so carry $\sum_t \gamma^t$ inside onto that weight, which collects into the discounted visitation measure $\eta(s) := \sum_{t \geq 0} \gamma^t \mathbb{P}(S_t=s)$ (the same object the recursive route built by unrolling, read in full there):

$$\begin{aligned} \nabla_{\theta} J(\theta) &= \sum_s \underbrace{\left[\sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t=s) \right]}_{\eta(s)} \sum_a \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s) \\ &= \sum_s \eta(s) \sum_a \nabla_{\theta} \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a), \end{aligned}$$

the last equality by $\pi_{\theta} \nabla_{\theta} \log \pi_{\theta} = \nabla_{\theta} \pi_{\theta}$. With the normalized visitation $d^{\pi_{\theta}, \rho_0}(s) = (1 - \gamma) \eta(s)$ this is exactly the theorem the recursive route produced:

$$\begin{aligned} \nabla_{\theta} J(\theta) &= \frac{1}{1 - \gamma} \sum_s d^{\pi_{\theta}, \rho_0}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \\ &= \frac{1}{1 - \gamma} \mathbb{E}_{S \sim d^{\pi_{\theta}, \rho_0}, A \sim \pi_{\theta}(\cdot | S)} [\nabla_{\theta} \log \pi_{\theta}(A | S) Q^{\pi_{\theta}}(S, A)]. \end{aligned}$$

For a genuinely episodic task one reads this with $\gamma = 1$: $\eta(s)$ is then the expected number of visits to the transient state s before absorption (finite, since the episode ends almost surely) and the front constant becomes the expected episode length. The visitation here is not predicted but lived: the states S_t in the expectation are simply the ones the sampled trajectory passed through, so running the policy draws $S \sim d^{\pi_{\theta}, \rho_0}$ for free. With the score analytic and $Q^{\pi_{\theta}}$ estimated from the sampled returns, this line is where REINFORCE and the actor-critic methods begin.